

Row 8, rebuilt from scratch

Is there any other UV divergence? — and the result in $\overline{\text{MS}}$

$$S \equiv p^+ + k^+, \quad r \equiv y' - y, \quad \kappa \equiv \frac{k \cdot r}{k^+}, \quad P_V \equiv V - k^+, \quad d = 2 - 2\epsilon$$

Answer. No — there is no other UV divergence. Redone from scratch: the tensor contraction by explicit index sums in general d (§1), all 26 collapsing regions with the Wilson lines actually present (§2), and the large-distance end of *every* variable done analytically with an exact Bessel identity (§3). The only divergence in this row is the one already found: the logarithmic **infrared** end of the z integral. **But there is a trap**, and it is exactly the one your question walks into: the standard way of doing that z integral *does* generate $\frac{1}{\epsilon_{\text{UV}}}$ poles in intermediate steps, which cancel (§4). If you $\overline{\text{MS}}$ -subtract the surviving $\frac{1}{\epsilon}$ you are subtracting an **infrared** pole. §5 gives the honest $\overline{\text{MS}}$ form, and it corrects the scheme constant I quoted last time.

1. The contraction, redone with an index engine

Last time I used a hand-worked table. Here the two brackets are built as explicit rank-4 arrays in d dimensions, contracted over i, j , and the result least-squares decomposed onto the three available structures, for $d = 2 \dots 7$ with $a = S/p^+ = 1.7$, $b = S/k^+ = 2.9$:

d	coef of $\delta_{km}\delta_{k'm'}$	coef of $\delta_{mm'}\delta_{kk'}$	coef of $\delta_{km'}\delta_{k'm}$	fit residual
2	-7.2000000000	11.3000000000	9.8600000000	$\sim 10^{-15}$
4	-5.2000000000	11.3000000000	9.8600000000	$\sim 10^{-15}$
6	-3.2000000000	11.3000000000	9.8600000000	$\sim 10^{-15}$
7	-2.2000000000	11.3000000000	9.8600000000	$\sim 10^{-14}$

Fitted slope in d : exactly 1. So, independently of anything I did before,

$$\text{coef}_1 = d - 2a - 2b, \quad \text{coef}_2 = a^2 + b^2, \quad \text{coef}_3 = 2ab,$$

and multiplying by the $\frac{p^+}{S^2}$ from the deltas reproduces your bracket with symbolic difference 0 in all three slots. **The point that matters for §5:** the d -dependence sits in the $\delta_{km}\delta_{k'm'}$ coefficient *and nowhere else*, which is what makes $d_\perp$ a live scheme choice for that one term only.

2. All 26 collapsing regions, with the Wilson lines in

The previous scan used bare kernels. Here the colour factor $[U(y') - U(x')][U(y) - U(x)]$ is carried along with a smooth non-trivial adjoint field $U^{ab}(p)$ built from $SU(3)$ generators. Every subset of $\{x, x', y, y', z\}$ of size ≥ 2 , collapsed at scale ρ against the measure $\rho^{2(n-1)}$:

region	$\rho = 10^{-2}$	10^{-4}	10^{-6}	power
$\{y, y'\}$	2.084e-3	2.084e-7	2.084e-11	ρ^2
$\{x, y\}$	1.004e-2	1.004e-6	1.004e-10	ρ^2
$\{x', y'\}$	4.443e-3	4.444e-7	4.444e-11	ρ^2
$\{y, z\}$	2.168e-1	2.168e-3	2.168e-5	ρ^1
$\{y', z\}$	2.146e-1	2.146e-3	2.146e-5	ρ^1
$\{y, y', z\}$	1.048e-3	1.048e-7	1.048e-11	ρ^2
$\{x, x', y, y'\}$	2.115e-10	2.115e-22	2.115e-34	ρ^6
$\{x, x', y, y', z\}$	7.230e-12	7.231e-24	7.231e-36	ρ^6

... and the other 18 subsets, all falling. Full table in the results file.

All 26 convergent. Note $\{x, y\}$ and $\{x', y'\}$ are now ρ^2 , not ρ^1 : with the Wilson lines present $U(y) - U(x) \rightarrow 0$ linearly, so those regions are *better* than the bare count. Including the colour can only help. No UV anywhere.

3. The large-distance end, one variable at a time

This is where the previous write-up was thin, so here it is analytically. For a variable swept to $|p| = R$ the radial integral converges iff the angular average $A(R)$ times R^2 falls with R . The exact 2D identity (verified numerically to 7 digits, $x = 10 \dots 10^5$):

$$\left\langle \hat{n}^m \hat{n}^k e^{ix \hat{n} \cdot \hat{k}} \right\rangle_{\theta} = \frac{J_1(x)}{x} \delta^{mk} + \left(J_0(x) - \frac{2J_1(x)}{x} \right) \hat{k}^m \hat{k}^k.$$

variable	kernels	phase	behaviour of $ A(R) R^2$	verdict
z	two	none	$x = 0: \frac{J_1}{x} \rightarrow \frac{1}{2}$, so $\langle \hat{n}^m \hat{n}^{m'} \rangle = \frac{\delta^{mm'}}{2} \neq 0 \rightarrow \text{const}$	log divergent
y, y'	two	$e^{\pm ik \cdot y} S/k^+$	$x = k RS/k^+: J_0(x) \sim x^{-1/2} \rightarrow R^{-1/2}$	convergent
x, x'	one	none	first two angular moments vanish identically $\rightarrow R^{-1}$	convergent

For x the cancellation is worth writing out, because it is not obvious:

$$\frac{(x-y)^k}{(x-y)^2} = \frac{\hat{n}^k}{R} + \frac{2\hat{n}^k(\hat{n} \cdot y) - y^k}{R^2} + O(R^{-3}), \quad \langle \hat{n}^k \rangle = 0, \quad \left\langle 2\hat{n}^k(\hat{n} \cdot y) - y^k \right\rangle = 2 \cdot \frac{y^k}{2} - y^k = 0.$$

Numerically, z flatlines at $|A(R)|R^2 = 92.9151$ from $R = 10^2$ to 10^5 — six digits, no drift. Everything else falls.

So the mechanism is sharp: y and y' are saved by the phase, x and x' by the vanishing of the first two angular moments of a single WW kernel. z has neither — it is the one variable the phase never touches, and the colour reduction $U^{d'b}(z)U^{db}(z) = \delta^{d'd}$ removed the last $U(z)$ that could have cut it off. The entire divergence of this row is that one factor of $\frac{1}{2}$ in $\langle \hat{n}^m \hat{n}^{m'} \rangle$.

4. The trap: where $\frac{1}{\epsilon_{UV}}$ does appear

First, the direct evidence that the physical integrand has no UV. Excise discs of radius a around y and y' from the trace integral (outer radius fixed at $R = 400$):

a	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
trace integral	36.5132	36.5125	36.5160	36.4654	36.4654

Total drift over four decades: 0.05, which is grid noise. A logarithmic UV divergence would have moved it by $\pi \log(10^4)^2 \approx 58$. Growing the outer radius instead ($a = 10^{-6}$) tracks $\pi \log \frac{R^2}{r^2}$ to 2×10^{-2} at $R = 50 \dots 3200$. The divergence is infrared, full stop.

Now the trap. The trace is almost always evaluated by writing $u \cdot v = \frac{1}{2}(u^2 + v^2 - r^2)$:

$$\int d^d z \frac{u \cdot v}{u^2 v^2} = \frac{1}{2} \int \frac{d^d z}{v^2} + \frac{1}{2} \int \frac{d^d z}{u^2} - \frac{r^2}{2} G(1, 1), \quad G(1, 1) = \int \frac{d^d z}{u^2 v^2}.$$

Both pieces on the right are UV divergent.

- The tadpole $\int \frac{d^d z}{u^2}$ is *scaleless*. It is set to zero, but only because

$$\int \frac{d^d z}{(z-y)^2} = \pi \left[\frac{1}{\epsilon_{IR}} - \frac{1}{\epsilon_{UV}} \right],$$

i.e. a genuine short-distance divergence at $z \rightarrow y$ cancelled against a genuine large-distance one. It is not finite; it is two divergences set equal.

- The bubble $G(1, 1)$ goes as $\rho^{d-3} d\rho$ at $z \rightarrow y$, so it is logarithmically UV divergent at $d = 2$. Its $\frac{1}{\epsilon}$ is $\frac{1}{\epsilon_{UV}}$.

The spurious $\frac{1}{\epsilon_{UV}}$'s cancel between the tadpoles and the bubble, and the $\frac{1}{\epsilon}$ that survives is $\frac{1}{\epsilon_{IR}}$ — which is what the numbers above measure directly. **If you hand that $\frac{1}{\epsilon}$ to an $\overline{MS}$ subtraction as though it were ultraviolet, you have absorbed an infrared divergence into your coupling.** It must cancel against the companion rows instead.

5. $\overline{\text{MS}}$

With the standard $\overline{\text{MS}}$ factor riding on $g^2 \mu^{2\epsilon}$, and $C(\epsilon) = \pi^{1-\epsilon} \frac{\Gamma(1+\epsilon)\Gamma(1-\epsilon)^2}{\Gamma(2-2\epsilon)} = \pi e^{-\epsilon(\log \pi + \gamma_E - 2)}$:

$$\left(\frac{e^{\gamma_E}}{4\pi}\right)^\epsilon C(\epsilon) (r^2 \mu^2)^{-\epsilon} \frac{1}{\epsilon} = \pi \left[\frac{1}{\epsilon} + 2 - \log(4\pi^2 \mu^2 r^2) \right] \equiv \pi [\mathcal{L} + 2], \quad \boxed{\mathcal{L} = \frac{1}{\epsilon} - \log(4\pi^2 \mu^2 r^2)}.$$

This corrects the constant I gave you last time. I wrote $\bar{\mu}^2 = \pi e^{\gamma_E} \mu^2$, which is a position–space convention, not $\overline{\text{MS}}$. Proper $\overline{\text{MS}}$ gives $4\pi^2 \mu^2$; the two differ by $\log \frac{4\pi}{e^{\gamma_E}}$, precisely the $\overline{\text{MS}}$ factor. The $4\pi^2$ rather than a bare μ^2 is there because $\int d^d z$ carries no $(2\pi)^{-d}$ — if your convention puts one there, drop it accordingly. Everything structural below is unchanged.

5.1 Scheme choice for $d_\perp$

By §1 the only structure carrying $d_\perp$ is $\delta_{km} \delta_{k'm'}$, and it multiplies $\frac{1}{2\epsilon}$ — so the choice is finite but real:

scheme	$d_\perp$	$\frac{d_\perp}{2\epsilon}$	$\delta_{km} \delta_{k'm'}$ contribution
CDR (= $\overline{\text{MS}}$)	$2 - 2\epsilon$	$\frac{1}{\epsilon} - 1$	$\frac{p^+}{S^2} [(\mathcal{L} + 1)\mathcal{X} - 2\mathcal{R}]$
DRED / FDH	2	$\frac{1}{\epsilon}$	$\frac{p^+}{S^2} [(\mathcal{L} + 2)\mathcal{X} - 2\mathcal{R}]$

Difference = $\frac{p^+ \mathcal{X}}{(k^+ + p^+)^2}$. Since you asked for $\overline{\text{MS}}$, use the first line — $d_\perp = 2 - 2\epsilon$ must be kept everywhere, including inside δ_{ii} .

5.2 The z -integrated bracket in $\overline{\text{MS}}$

With $\mathcal{X} = \frac{(x-y) \cdot (x'-y')}{(x-y)^2 (x'-y')^2}$ and $\mathcal{R} = \frac{[r \cdot (x-y)][r \cdot (x'-y')]}{r^2 (x-y)^2 (x'-y')^2}$, the $+2$ from $\mathcal{L} + 2$ cancels against the -2 of $\frac{1-2\epsilon}{\epsilon}$ in the $\delta_{mm'}$ term, and the $\frac{2}{k^+}$ term dies because $T^{mm'}$ is symmetric:

$$\pi \left\{ \frac{\mathcal{L} \mathcal{X}}{p^+} + \frac{p^+}{k^{+2}} \mathcal{L} \mathcal{X} + \frac{p^+}{(k^+ + p^+)^2} [(\mathcal{L} + 1)\mathcal{X} - 2\mathcal{R}] \right\}$$

6. The result in $\overline{\text{MS}}$

$$\left. \frac{d^3 N^{\text{NLO}}}{d^3 k} \right|_{\text{cc}} = \frac{1}{(2\pi)^3} \frac{g^4 N_c}{4\pi^4} \frac{1}{2k^+} \int_{x, x', y, y'} e^{-ik \cdot (y' - y)} \left\{ \mathcal{L} \mathcal{X} \left[\mathcal{I}_1 + \frac{\mathcal{I}_2}{k^{+2}} \right] + [(\mathcal{L} + 1)\mathcal{X} - 2\mathcal{R}] \mathcal{I}_3 \right\} \\ \times [U^{e'c}(y') - U^{e'c}(x')] [U^{ec}(y) - U^{ec}(x)] \rho^{e'}(x') \rho^e(x)$$

$\mathcal{L} = \frac{1}{\epsilon} - \log(4\pi^2 \mu^2 r^2)$, $r = y' - y$, $\kappa = \frac{k \cdot r}{k^+}$, $P_V = \vee - k^+$, and

$$\mathcal{I}_1 = \underbrace{\log \frac{\vee - k^+}{\Lambda}}_{\text{rapidity}} + \underbrace{\int_0^{P_V} \frac{dp^+}{p^+} [e^{-i\kappa p^+} - 1]}_{[1/p^+]_+} = \log \frac{P_V}{\Lambda} - \text{Cin}(|\kappa| P_V) - i \text{sgn}(\kappa) \text{Si}(|\kappa| P_V), \\ \mathcal{I}_2 = \frac{1 - e^{-i\kappa P_V} (1 + i\kappa P_V)}{-\kappa^2}, \quad \mathcal{I}_3 = e^{i\kappa k^+} \left[(1 + i\kappa k^+) (\text{Ei}(-i\kappa \vee) - \text{Ei}(-i\kappa k^+)) + k^+ \left(\frac{e^{-i\kappa \vee}}{\vee} - \frac{e^{-i\kappa k^+}}{k^+} \right) \right].$$

6.1 What μ means here

Matching to the cutoff form $\frac{\pi}{2} \left(\log \frac{R^2}{r^2} + 1 \right)$ gives

$$R = \frac{e^{1/2}}{2\pi\mu}.$$

R is the **outer** radius of the z integration. So μ here is an **infrared** scale, not a renormalisation scale — large μ means a *small* z -plane. Running the coupling with it, or absorbing $\mathcal{L}$ into $\alpha_s(\mu)$, would be wrong. The $\frac{1}{\epsilon}$ has to cancel against the rows where a $U(z)$ survives to cut the same integral off at the target size.

6.2 Two things unchanged, one changed

- **Unchanged:** no UV anywhere; the $\frac{2}{k^+}$ term cancels identically; the pole is IR and rides on the LO structure; the $+$ prescription on $\mathcal{I}_1$ is legitimate because the z integral was made finite *first* and the $p^+ \rightarrow 0$ subtraction keeps the LO phase $e^{-ik \cdot (y' - y)}$.
- **Changed:** the scheme constant inside the log, $\pi e^{\gamma_E} \mu^2 \rightarrow 4\pi^2 \mu^2$ (§5), and the $d_\perp$ scheme is now pinned to CDR, giving $(\mathcal{L} + 1)$ rather than $(\mathcal{L} + 2)$ (§5.1).
- **Caveat worth checking separately:** if you then take an MV/Gaussian average $\langle \rho^a(x) \rho^b(x') \rangle \propto \delta^{ab} \delta^{(2)}(x - x')$, the $x = x'$ identification turns $\mathcal{X}$ into the dipole kernel $\mathcal{K}(y, y'; x)$, and the x integral then has the same unprotected structure the z integral had — a *second* infrared log. That is a statement about the averaging, not about this row, but it will double the pole.

zint/src/row8_redo.py (§1–2), row8_asym.py, row8_bessel.py (§3), row8_uvswEEP.py (§2, all three stages), row8_msbar.py (§4–5) — every number above.